1 A student is investigating how the boiling point of a salt solution varies with pressure.

It is suggested that the relationship between the Celsius temperature θ at which the water of the solution starts to boil and the air pressure P is

$$\theta = k\sigma P^q$$

where σ is the density of the solution and k and q are constants.

Design a laboratory experiment to test the relationship between θ and P.
Explain how your results could be used to determine values for k and q.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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